Fiscal Year:	FY 2008	**Task Last Updated:**	FY 05/15/2008
PI Name:	Wang, Ya M.D., Ph.D.		
Project Title:	Checkpoint Reduced Cell Sensitivity to High Energy Particles-Induced Killing		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	30322	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
Start Date:	07/01/2003	**End Date:**	06/30/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes then, per JSC/A. Chu-ARC (jvp 4/2009) NOTE: Received NCE to 6/30/2008 per A. Chu (6/07) NOTE: End date changed to 6/30/07 per A. Chu (5/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1628		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This application is in response to the NASA Program NRA 02-OBPR-02 and addresses research related to both Bioastronautics and Fundamental Biology. Radiation-induced DNA double strand breaks (DSBs) are a severe threat to genomic integrity that determines radiosensitivity (short term effect) and carcinogenesis (long term effect). Two major DNA DSB repair pathways exist in eukaryotic cells, homologous recombination repair (HRR) and non-homologous end joining (NHEJ). After induction of DSBs, proliferating cells actively slow down cell cycle progression via checkpoint activation to provide time for repair. However, it is not known whether checkpoint activation benefits NHEJ and HRR equally, or whether it is specifically adapted to assist just one of these processes, therefore reducing DSBs-induced genomic instability. The long-term objective of the present proposal is to elucidate the molecular mechanisms by which checkpoint activation facilitates DNA repair and therefore reduces the high-energy (HZE) particles-induced genomic integrity in mammalian cells. In this proposal, the central hypothesis is that CHK1-regulated checkpoints, in an ATR-dependent but ATM-independent manner, facilitate HRR but not NHEJ, thus affecting HZE particles-induced killing of mammalian cells. The preliminary results from sparsely ionizing radiation provide evidence to support this hypothesis and form the foundation for the experimental design. Three Specific Aims are pursued: Specific aim 1: Determine whether ATR, the upstream kinase of CHK1, affects HZE particles-induced killing by facilitating HRR. We will examine HRR (by using the GFP-I-SceI restriction enzyme system), NHEJ (by using asymmetric field inversion gel electrophoresis, AFIGE), CHK1 activity (by using kinase assays) and the effect of CHK1 siRNA on HZE particles-induced killing (by using clonogenic assay) in ATR kinase dead (ATR-kd) human fibroblasts. Specific aim 2: Determine whether the protective effect of the CHK1-regulated checkpoint on HZE particles-induced killing is mediated by facilitating HRR. We will transfect the CHK1-kd constructs combined with Tet-OnTM systems to A1-5 cells, a rat embryo fibroblast line in which CHK1 is highly expressed. We will examine HRR, NHEJ, CHK1 activity and HZE particles-induced killing in CHK1-kd cell lines. Specific aim 3: Determine whether the effect of CHK1 on HZE particles-induced killing is ATM independent. We will transfect the CHK1-kd constructs combined with Tet-OnTM systems to AT5BIVA (ATM-/-) human fibroblast cells. We will test HZE particles-induced killing, HRR, NHEJ, CHK1 activity and checkpoint activation in the CHK1-kd AT cell lines following HZE particles radiation. This approach will clarify the relationships among ATR/CHK1-regulated checkpoint activation, NHEJ and HRR. Such information will provide means for protecting mammals from HZE particles-induced killing in a short term and from HZE particles-induced carcinogenesis in a long term by activating a particular pathway (ATM or ATR-regulated checkpoints, NHEJ or HRR) depending on the genetic background of the mammals. The proposal addresses a basic research topic related to the protection of individuals from the hazards of ionizing radiation in outer space. The proposed basic research is within the framework defined by the Space Radiation Health Program, and attempts to establish the scientific basis for radiation protection of humans engaged in the exploration of outer space. It is a ground-based program that studies the hypothesis that CHK1-regulated checkpoints, in an ATR-dependent but ATM-independent manner, facilitate HRR but not NHEJ, thus affecting HZE particles-induced killing of mammalian cells. We believe that our results from this proposal will clarify the relationships among ATR/CHK1-regulated checkpoint activation, NHEJ and HRR. Once we understand the relationships among checkpoint activation, HRR and NHEJ, we will be able to protect cells from HZE particles-induced killing by activating the particular pathway (ATM- or ATR/CHK-regulated checkpoint, HRR or NHEJ) of the cells with different genetic backgrounds. We expect that these results will have the potential for reducing the HZE particles-induced mutagenesis and carcinogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results from these experiments will clarify the role of the ATR/CHK1 pathway, the most important checkpoint proteins in mammalian cells, in protecting cells from ionizing radiation (IR)-induced killing.
	Description: This project (02-OBPR-02-0000-0037) that was started on July 1, 2003 has been successfully finished. Radiation-induced DNA double strand breaks (DSBs) are a severe threat to genomic integrity that determines radiosensitivity (short-term effect) and carcinogenesis (long-term effect). Two major DNA DSB repair pathways exist in eukaryotic cells, homologous recombination repair (HRR) and non-homologous end joining (NHEJ). After induction of DSBs, proliferating cells actively slow down cell cycle progression via checkpoint activation to provide time for repair. However, it is not known whether checkpoint activation benefits NHEJ and HRR equally, or whether it is specifically adapted to assist just one of these processes therefore reducing DSBs-induced genomic instability. The long-term objective of the present proposal is to elucidate the molecular mechanisms by which checkpoint activation facilitates DNA repair and therefore reduces the high-energy (HZE) particles-induced genomic integrity in mammalian cells. In this proposal, the central hypothesis is that CHK1-regulated checkpoints, in an ATR-dependent but ATM-independent manner, facilitate HRR but not NHEJ, thus affecting HZE particles-induced killing of mammalian cells. The preliminary results from sparsely ionizing radiation provide evidence to support this hypothesis and form the foundation for the experimental design. Three Specific Aims are pursued: Specific aim 1: Determine whether ATR, the upstream kinase of CHK1, affects HZE particles-induced killing by facilitating HRR. We will examine HRR (by using the GFP-I-SceI restriction enzyme system), NHEJ (by using asymmetric field inversion gel electrophoresis, AFIGE), CHK1 phosphorylation (by using Western blot) and the effect of CHK1 siRNA on HZE particles-induced killing (by using clonogenic assay) in ATR kinase dead (ATR-kd) human fibroblasts. Specific aim 2: Determine whether the protective effect of the CHK1-regulated checkpoint on HZE particles-induced killing is mediated by facilitating HRR. We will transfect the CHK1-kd constructs combined with Tet-OnTM systems to GM847, the parental cell line of ATR-kd. We will examine HRR, NHEJ, CHK1 activity and HZE particles-induced killing in CHK1-kd cell lines. Specific aim 3: Determine whether the effect of CHK1 on HZE particles-induced killing is ATM independent. We will transfect the CHK1-kd constructs combined with Tet-OnTM systems to AT5BIVA (ATM-/-) human fibroblast cells. We will test HZE particles-induced killing, HRR, NHEJ, CHK1 phosphorylation and checkpoint activation in the CHK1-kd AT cell lines following HZE particles radiation. So far, with 6 sessions of experiment by using high linear energy transfer (LET) ionizing radiation (IR) in Brookhaven National Laboratory (BNL), we have followed the proposal to carry out all the experiments as described in our specific aims. We have reached the goal of this planned proposal. The products from this proposal are reflected in our publications as follows:

Task Progress:	Publications: 1. Wang, X., Khadpe, J., Hu, B., Iliakis, G., and Wang, Y. An over-activated ATR/CHK1 pathway is responsible for the prolonged G2 accumulation in irradiated AT cells. J Biol. Chem., 278: 30869-30874, 2003. 2. Wang, H.Y., Wang, H.C., Powell, S.N., Iliakis, G. and Wang, Y. ATR affecting cell radiosensitivity is dependent on homologous recombination repair but independent of non-homologous end joining. Cancer Res. 64: 7139-7143, 2004. 3. Wang, H.Y., Hu, B., Liu, R. and Wang, Y. CHK1 Affecting Cell Radiosensitivity is Independent of Non-homologous End Joining. Cell Cycle 4: 300-303, 2005. 4. Hu, B., Wang, H.Y., Wang, X., Lu, H-R, Huang, C., Powell, S.N., Huebner, K. and Wang, Y. Fhit and CHK1 Have Opposing Effects on Homologous Recombination Repair. Cancer Research, 65: 8613-8616, 2005. 5. Wang, X., Hu, B., Weiss , R.S. and Wang, Y. The Effect of Hus1 on Ionizing Radiation Sensitivity is Associated with Homologous Recombination Repair but is Independent of Non-Homologous End-joining. Oncogene, 25: 1980-1983, 2006. 6. Lu, H-R, Wang, X and Wang, Y. A Stronger DNA Damage-induced G2 Checkpoint due to Over-activated CHK1 in the Absence of PARP-1. Cell Cycle, 5: 2364-2370, 2006. 7. Wang, H., Wang, X. Zhang, PY and Wang, Y. The Ku-dependent non-homologous end-joining but not other repair pathway is inhibited by high-linear energy transfer ionizing radiation. DNA Repair, 7: 725-733, 2008. 8. Wang, H. and Wang, Y. Checkpoint response plays a more protective role in HZE particle-irradiated cells than in X ray-irradiated cells. Cell Cycle. Accepted, 2008. Presentations: 1. Wang, X., Khadpe, J., Hu, B., Iliakis, G., and Wang, Y. An over-activated ATR/CHK1 pathway is responsible for the prolonged G2 accumulation in irradiated AT cells. 12th International Congress of Radiation Research, Brisbane, Australia, August 2003. 2. Wang, X., Guan, J., Hu, B., Weiss, R.B., Iliakis, G. and Wang, Y. Involvement of Hus1 in the chain elongation step of DNA replication after exposure to camptothecin or ionizing radiation. AACR Radiation Biology and Cancer: from molecular research to the clinic. Dana Point, CA, February 2004. 3. Wang, H.Y., Hu, B., Liu, R. and Wang, Y. CHK1 Affecting Cell Radiosensitivity is Independent of Non-homologous End Joining. 3rd International Workshop on Space Radiation Research, Port Jefferson, New York, May 2004. 4. Wang, X. and Wang, Y. MEPE/OF45 is a Substrate of CHK1 Kinase to Respond to DNA Damage. 53rd Annual Meeting of the Radiation Research Society, Philadelphia, PA, November 5-8, 2006. 5. Wang, H., Wang, X., Tekeda, S. and Wang, Y. Ku dependent NHEJ pathway inhibits camptothecin-induced S phase checkpoint. 53rd Annual Meeting of the Radiation Research Society, Philadelphia, PA, November 5-8, 2006. 6. Lu, H-R and Wang, Y. MEPE/OF45 Affects the Cellular Sensitivity to Ionizing Radiation. 53rd Annual Meeting of the Radiation Research Society, Philadelphia, PA, November 5-8, 2006. 7. Wang, H. Wang, X. and Wang, Y. The Ku-dependent Non-homologous End-joining but not Other Repair Pathway is Inhibited by High-linear Energy Transfer Ionizing Radiation. NASA Models of Space Radiation Risks Workshop, Dallas, March 6-7, 2007. 8. Wang, H. and Wang, Y. ATM is not directly involved in Ku-dependent non-homologous end-joining DNA repair. 18th Annual NASA Space Radiation Investigators' Workshop, Rohnert Park, California, July 13-15, 2007.
Bibliography Type:	Description: (Last Updated: 07/07/2021)
Abstracts for Journals and Proceedings	Wang H, Wang Y. "ATM is not directly involved in Ku-dependent non-homologous end-joining DNA repair." 18th Annual NASA Space Radiation Investigators' Workshop, Rohnert Park, California, July 13-15, 2007. Proceedings, 18th Annual NASA Space Radiation Investigators' Workshop, Rohnert Park, California, July 13-15, 2007. , Jul-2007
Articles in Peer-reviewed Journals	Wang H, Wang X, Zhang P, Wang Y. "The Ku-dependent non-homologous end-joining but not other repair pathway is inhibited by high-linear energy transfer ionizing radiation." DNA Repair (Amst). 2008 May 3;7(5):725-33. Epub 2008 Mar 5. PMID: 18325854 , May-2008
Articles in Peer-reviewed Journals	Wang H, Wang Y. "Checkpoint response plays a more protective role in HZE particle-irradiated cells than in X ray-irradiated cells." Cell Cycle. Accepted, 2008. Forthcoming in issue: 2008 Jul 15;7(14). , Jul-2008